SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1014	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Arunprasath R	Reg. No.:	192524077
Date:	10/10/2026	Duration:	60 minutes

Code of Conduct

I Arunprasath R (192524077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: Arunprasath R

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders o Define the project's vision, objectives, and identify all key stakeholders involved.
2. Create Epics and User Stories o Break down the requirements into Epics and formulate well-structured User Stories using the format: As a <user>, I want <feature>, so that <benefit>.

3. Define Acceptance Criteria o Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. Prioritize the Product Backlog o Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. Plan the Sprint Backlog o Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. Estimate Effort Using Story Points o Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. Present the Sprint Goal and Expected Deliverables o Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

QUESTION:

Intelligent Academic Timetable Conflict Resolution Platform

Industry Context

Universities must schedule hundreds of courses, faculty members, classrooms, laboratories, and student batches. Manual timetable preparation can create conflicts and inefficient room utilization.

Problem Statement

Design and develop an intelligent timetable platform that automatically detects scheduling conflicts and generates feasible timetables based on faculty availability, classroom capacity, laboratory requirements, course combinations, and student batches.

Objectives

- Manage academic scheduling constraints.
- Detect faculty conflicts.
- Detect classroom conflicts.
- Handle laboratory requirements.
- Optimize room utilization.
- Support timetable modifications.

Expected Outcomes

- Conflict-free timetables.
- Reduced administrative effort.
- Better classroom utilization.
- Faster timetable generation.
- Improved academic scheduling.

ANS:

1. Project Vision and Stakeholders:

Project Vision

To develop an intelligent platform that automatically generates academic timetables by considering faculty availability, classroom capacity, laboratory requirements, and student batch schedules while minimizing conflicts.

Objectives:

- Manage courses, faculty, classrooms, and student batches.
- Detect faculty, classroom, and student batch conflicts.
- Generate feasible academic timetables automatically.
- Improve classroom and laboratory utilization.
- Allow timetable modifications.

Stakeholders:

1. University Administrator: Manages academic resources.
2. Timetable Coordinator: Creates and modifies timetables.
3. Faculty Members: Provide availability and view schedules.
4. Students: View their class timetables.
5. Department Heads: Review and approve timetables.
6. Developers and Testers: Develop and validate the system.

2.Create Epics and User Stories :

Agile development emphasizes breaking down large requirements into **Epics** (broad features) and **User Stories** (specific, actionable needs). For this project, the following epics and stories capture the essential functionality:

Epic 1: Faculty Scheduling
Description: Manage faculty availability and teaching preferences to avoid conflicts.

- **User Story 1.1:** As a faculty member, I want to input my availability, so that the system avoids scheduling classes during my unavailable hours.
- **User Story 1.2:** As a faculty member, I want to specify preferred teaching slots, so that my schedule aligns with my personal and academic commitments.

Epic 2: Classroom & Laboratory Allocation
Description: Ensure classrooms and labs are assigned based on capacity and requirements.

- **User Story 2.1:** As an administrator, I want the system to assign classrooms based on student capacity, so that room utilization is optimized.
- **User Story 2.2:** As a lab coordinator, I want to allocate laboratories with required equipment, so that students can complete practical sessions without disruption.
- **User Story 2.3:** As an administrator, I want to view available classrooms in real time, so that I can quickly resolve scheduling conflicts.

Epic 3: Conflict Detection
Description: Automatically identify and resolve scheduling conflicts.

- **User Story 3.1:** As an administrator, I want the system to detect overlapping schedules, so that conflicts are resolved before publishing.
- **User Story 3.2:** As a faculty member, I want to be notified if I am scheduled for two classes at the same time, so that adjustments can be made immediately.
- **User Story 3.3:** As a student, I want the system to prevent overlapping courses in my timetable, so that I can attend all required sessions.

Epic 4: Timetable Generation & Modification
Description: Generate conflict-free timetables and allow dynamic modifications.

- **User Story 4.1:** As a student, I want to view a conflict-free timetable, so that I can plan my academic schedule effectively.
- **User Story 4.2:** As an administrator, I want to modify timetables dynamically, so that last-minute changes can be accommodated.
- **User Story 4.3:** As a faculty member, I want to request timetable changes, so that my schedule reflects updated availability.

Epic 5: Optimization and Reporting

Description: Enhance efficiency and provide insights into scheduling.

- **User Story 5.1:** As an administrator, I want the system to optimize room utilization, so that resources are used efficiently.
- **User Story 5.2:** As management, I want to generate reports on timetable efficiency, so that I can evaluate academic resource usage

3.Define Acceptance Criteria: US01: Course Management:

- Given the administrator enters valid course details,
- When the administrator saves the information,
- Then the course must be stored successfully.
- Duplicate course IDs must be rejected. US02: Faculty Management:
- Given valid faculty details,
- When the administrator registers a faculty member,
- Then the faculty details must be saved successfully.
- Duplicate faculty IDs must not be allowed.

US03: Room Management:

- Given valid classroom or laboratory details,
- When the administrator registers a room,
- Then the room capacity and type must be stored.
- Room capacity must be greater than zero.

US04: Conflict Detection:

- Given a faculty member is assigned to multiple classes,
- When the assignments overlap in time,
- Then the system must identify and display the conflict.

US05: Timetable Generation:

- Given valid academic data and scheduling constraints,
- When the coordinator requests timetable generation,
- Then the system must generate a feasible timetable if one exists.
- Faculty, classroom, and student batch conflicts must be avoided.

US06: Room Optimization:

- Given a timetable has been generated,
- When the system evaluates room assignments,
- Then it must identify opportunities to improve room utilization without introducing conflicts.

US07: Timetable Modification:

- Given an existing timetable,
- When the coordinator changes a class assignment,
- Then the system must validate the change and identify any conflicts.

US08: Timetable Access:

- Given a timetable has been published,
- When a student selects their batch,
- Then the system must display the corresponding timetable.

4.Prioritize the Product Backlog:

Priority	Story ID	User Story	Story Points
High	US01	Course Management	3
High	US02	Faculty Management	3
High	US03	Room Management	3
High	US04	Conflict Detection	5
High	US05	Timetable Generation	8
Medium	US06	Room Optimization	8
Medium	US07	Timetable Modification	5
Medium	US08	Timetable Access	3

Prioritization: Academic data management is prioritized because timetable generation depends on course, faculty, and room information. Conflict detection follows, ensuring that generated timetables satisfy mandatory scheduling constraints. Optimization and additional access features are implemented in later stages.

5.Plan the Sprint Backlog:
Sprint Number: Sprint 1
Duration: 2 Weeks
Sprint Objective: Develop academic data management and basic faculty conflict detection.

	Task ID	Development Task	Estimated Hours
	T1	Design the system data model	4
	T2	Develop course management	5
	T3	Develop faculty management	5
	T4	Develop classroom and laboratory management	5
	T5	Implement faculty conflict detection	8
	T6	Test and integrate the modules	5
	Total		32 hours

Selected User Stories:
- US01: Course Management

- US02: Faculty Management
- US03: Room Management □
- US04: Conflict Detection

Total Sprint 1 Estimate: 14 story points.
The estimates are preliminary and can be refined by the development team during sprint planning.

6. Sprint Goal and Expected Deliverables:

Sprint Goal:
To develop a working prototype that manages courses, faculty members, classrooms, and laboratories, and detects overlapping faculty assignments.
Expected Deliverables:
1. Course Management Module: Add and manage course details.
2. Faculty Management Module: Register faculty members and validate details.
3. Room Management Module: Maintain classroom and laboratory information.
4. Faculty Conflict Detection Module: Identify overlapping faculty assignments.
5. Testing Report: Verify data validation and conflict detection.
6. Working Prototype: Integrate and demonstrate the completed features.

Definition of Done:
- All acceptance criteria are satisfied.
- Development and integration are completed.
- Relevant test cases pass.
- No known critical defects remain in the completed features.
- The working prototype is demonstrated.

Sprint Goal: Deliver the foundational scheduling features: faculty availability, classroom allocation, and conflict detection to ensure the system can generate basic conflict-free timetables.

Expected Deliverables:

- Functional module for faculty availability input.
- Classroom allocation system with capacity validation.
- Conflict detection engine with alerts and suggestions.
- Documentation of sprint outcomes.

8. Agile Practices Applied

- **Sprint Planning:** Defined scope and backlog items for Sprint 1.
- **Daily Stand-ups:** Team alignment and progress tracking.
- **Sprint Review:** Demonstration of deliverables to stakeholders.
- **Sprint Retrospective:** Continuous improvement for next sprint.
- **Backlog Grooming:** Regular prioritization based on feedback.

9. Risks and Mitigation

- **Risk:** Faculty may not update availability promptly.
  - *Mitigation:* Automated reminders and notifications.
- **Risk:** Classroom capacity data may be outdated.
  - *Mitigation:* Integration with updated facility database.
- **Risk:** Algorithm complexity may delay delivery.
  - *Mitigation:* Incremental development and testing.

10. Future Enhancements

- Integration with student portals for personalized timetables.
- AI-driven optimization for resource allocation.
- Mobile application support.
- Advanced analytics for utilization trends.

Conclusion

The Agile sprint planning process divides the Intelligent Academic Timetable Conflict Resolution Platform into manageable tasks. Sprint 1 establishes the foundation for academic data management and faculty conflict detection, supporting future development of automatic timetable generation and utilization, and lays the foundation for advanced features like laboratory scheduling and timetable modifications.

The iterative Agile process ensures continuous stakeholder collaboration, adaptability to changes, and incremental delivery of value. Ultimately, the platform will transform academic scheduling into a streamlined, conflict-free, and intelligent process, aligning with the university's vision of efficiency and innovation.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.

Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25